College of Technology Galle

Course: Full Stack Software Development — Object Oriented Programming — Module K72T004M03 — NVQ Level 05

SEMESTER MODEL PAPER 06

Focus: Abstraction, interfaces, object relationships intro

Difficulty: Moderate

Duration: 3 Hours **Total Marks:** 100 **Questions:** 7 structured questions (answer any **5**) **Each question:** 20 marks

Instructions to Candidates

1. Answer any **five (5)** questions out of the seven provided.
2. Each question carries **20 marks**. Maximum mark for this paper is **100**.
3. Write answers in the answer booklet. Do **not** write answers on this question paper.
4. Where Java code is required, write clear, syntactically correct Java as taught in this module (VS Code / JDK).
5. Show workings / short explanations where asked. Manage your time (about 35 minutes per question).
6. Mobile phones and programmable devices are not allowed.

Question 1 [20 Marks]

Abstract classes.

a. What is an abstract class? Can you write `new AbstractType()` if the type is abstract? **(6 marks)**

b. Write abstract class `Employee` with an abstract method `pay()` and a concrete method `getName()`. **(8 marks)**

c. Write one concrete subclass that implements `pay()`. **(6 marks)**

Question 2 [20 Marks]

Interfaces.

a. What is an interface in Java? How does a class use one? **(6 marks)**

b. Write interface `Payable` with method `double amountDue()` and a class that implements it. **(8 marks)**

c. State two differences between an abstract class and an interface (as taught for this level). **(6 marks)**

Question 3 [20 Marks]

Association, aggregation, composition - concepts.

a. Explain association (uses-a) with a short Java-flavoured example. **(6 marks)**

b. Explain aggregation (has-a, shared lifetime) vs composition (part-of, strong ownership) with school/classroom examples. **(10 marks)**

c. Why does composition often create the part objects inside the whole's constructor? **(4 marks)**

Question 4 [20 Marks]

School project style.

a. Design classes `Teacher` and `Classroom` showing a clear relationship (state which relationship you chose). **(6 marks)**

b. Write Java outlines (fields + constructors + one method each) for your design. **(10 marks)**

c. How would you print all students in a classroom if the classroom aggregates a list of students? **(4 marks)**

Question 5 [20 Marks]

Combining encapsulation + inheritance.

a. Write `Account` with private balance, protected deposit helper or public deposit, and abstract `calcInterest()`. **(10 marks)**

b. Write `SavingsAccount` that extends it and implements interest calculation. **(6 marks)**

c. Explain how encapsulation still protects balance while inheritance reuses behaviour. **(4 marks)**

Question 6 [20 Marks]

```
interface Notifier {
    void send(String msg);
}
class EmailNotifier implements Notifier {
    public void send(String msg) { System.out.println("EMAIL: " + msg); }
}
```

Study the interface example.

a. What must every implementing class provide? **(4 marks)**

b. Write a `SmsNotifier` implementation. **(6 marks)**

c. Write a method `alert(Notifier n, String m)` that calls `n.send(m)`, and show calls with both implementations. **(10 marks)**

Question 7 [20 Marks]

Garbage collection & lifecycle (link to memory lesson).

a. Describe birth, life, and abandonment of an object in heap terms. **(6 marks)**

b. What is the Garbage Collector's job? Do you call `delete` in Java? **(6 marks)**

c. Show code that makes an object unreachable (e.g. assign `null` or reassign the only reference). **(4 marks)**

d. Why can large temporary objects in a loop cause memory pressure even with GC? **(4 marks)**

******* END OF QUESTION PAPER *******